Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Their Properties

You already know the properties of cubes, cuboids, pyramids and cones. Now let's look at two more solid shapes: the cylinder and the sphere. These are three-dimensional figures, and each has its own set of properties.

A cylinder is a solid shape with two flat circular faces and one curved surface joining them. Picture a tin of milk or a roll of paper towel. Its properties are:

It has two circular faces, one at the top and one at the bottom.
The two circular faces are equal in size — they are identical circles.
It has one curved surface that wraps around and joins the two circles.
It has two edges — each edge is the circle where a circular face meets the curved surface.
It has no vertices (corners).

So when you count the faces of a cylinder, you get two flat circular faces and one curved face. Altogether a cylinder has three faces: two flat circles and one curved surface.

Before we go further, see if you can name the properties of a sphere yourself. A sphere is a perfectly round solid shape — think of a ball, a football, or an orange. Write down what you think its properties are, then read on to check.

A sphere's properties are:

It has no faces.
It has no edges.
It has no vertices.
It has one curved surface that covers the whole shape.
Every point on its surface is the same distance from its centre.

Unlike the cylinder, which has flat circular faces, the sphere is curved all over — that is why it has no flat faces, no edges and no corners.

Let's compare the two shapes side by side.

Question: A solid cylinder has how many faces?

Solution: A cylinder has two flat circular faces (one at the top, one at the bottom) and one curved surface that joins them. So altogether it has three faces: 2 circular faces + 1 curved face = 3 faces.

Question: How many circular faces does a solid cylinder have?

Solution: The two flat faces are the circular ones. So a solid cylinder has two circular faces.

Question: How many faces does a sphere have?

Solution: A sphere is curved all over, so it has no flat faces at all — it has zero faces (just one curved surface covering the whole shape).

Practice

1. A solid cylinder has how many faces altogether?
2. How many circular faces does a solid cylinder have?
3. State the number of faces, edges and vertices of a sphere.
4. A cylinder has some vertices. How many?

Section 2: Finding Volume: Cubes, Cuboids and Prisms

Volume is the amount of space a solid takes up. For a cube or a cuboid, the volume is found by multiplying the three dimensions together: length × width × height.

A **cuboid** is a box with rectangular faces. Its volume is:

Volume = length × width × height

A **cube** is a special cuboid where all three edges are the same length, so its volume is simply:

Volume = side × side × side

You can think of it this way: the bottom face of a cuboid has area length × width, and the height tells you how many of those layers of unit cubes stack up. So the volume is just the area of the base multiplied by the height. This same idea works for any prism: a prism has a shape at the front and back that is the same all the way through, so

Volume of a prism = area of base × height

For a cuboid the base is a rectangle, so area of base = length × width, which gives length × width × height.

Example 1 — cuboid. Find the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.

Volume = length × width × height = 8 × 4 × 3 = 96

So the volume is 96 cm³.

Example 2 — cube. Find the volume of a cube with edges 4 cm.

Volume = side × side × side = 4 × 4 × 4 = 64

So the volume is 64 cm³.

Example 3 — prism. A prism has a triangular base. The base has area 12 cm² and the prism is 5 cm long. Find its volume.

Volume = area of base × height = 12 × 5 = 60

So the volume is 60 cm³.

Practice

1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube of 4 cm × 4 cm × 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m³ of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Their Properties

1. 3 faces (2 circular faces and 1 curved surface)

   The cylinder has a top circle, a bottom circle, and one curved surface wrapping around to join them. So 2 + 1 = 3 faces.
2. 2 circular faces

   The two flat faces of the cylinder are the circles at the top and at the bottom, so there are exactly two circular faces.
3. A sphere has 0 faces, 0 edges and 0 vertices. It has one curved surface covering the whole shape.

   A sphere is curved all over, with no flat faces and no corners, so it has no faces, no edges and no vertices.
4. 0 vertices

   A vertex is a corner where edges meet. A cylinder has only circular edges and no corners, so it has no vertices.

Finding Volume: Cubes, Cuboids and Prisms

1. 96 cm^3

   Volume = length × width × height = 8 × 4 × 3 = 96, so the volume is 96 cm^3.
2. 30 m^3

   Volume = 5 × 2 × 3 = 30, so the volume is 30 m^3.
3. 64 cm^3

   Volume = side × side × side = 4 × 4 × 4 = 64, so the volume is 64 cm^3.
4. 3 m

   Volume = length × width × depth, so 36 = 6 × 2 × depth = 12 × depth. Divide both sides by 12: depth = 36 ÷ 12 = 3. The water is 3 m deep.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-364e9e56	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____